Part-A

1) (B) 0

2) (C) 2

3) (A) 2

4) (A) 2

5) (A) 2

6) (C) 2

7) (C) 2

8) (B) 0

9) (B) 0

10) (C) 0

Part-B

11.2

▶ Forensic Anthropology : It is the branch of forensic science that deals with the study of body remains of (focused on Human beings) for the whole purpose of identification.

Ex- Working on a skull recovered from a crime scene, and identifying that to whom it belongs is under Forensic Anthropology.

▶ Significance of Anthropology in Forensic Science :

- It helps in the identification of certain individual.
- Can help in classifying people on basis of their

race, sex and age.

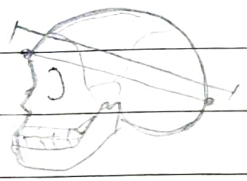
- With the help of Somatometry and Somatoscopy we can tell about the characteristics of any individual.

- We get to know about each and every bone in the Skeletal System:
  - Their use · Their formation · Their shape

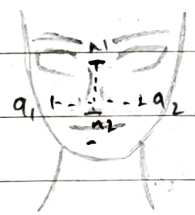
- Facial Reconstruction can also be done if we recover skull as an evidence or any part of skull.
- It becomes easy to classify b/w males and females if we ~~have~~ recover skull & mandible or pelvis.

➤ **Somatometry** : It is the process in which measurements of body parts like - Nose, ear, head, face is done for the purpose of personal identification.

- **Head** : The two most end points of skull are considered for the measurement of Head.



- **Nose** :



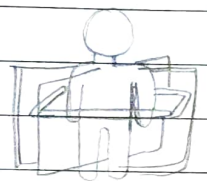
Nose of an individual is measured in two ways

- the length of nose from (n_1 to n_2)
- the width of nose from (a_1 to a_2)

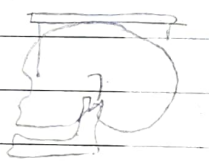
- **Face :** For measurement of face there is a way in which we measure distance b/w two zygomatic bones called (bizygomatic measurement) from (a_1 to a_2)



These are some of the measurements and few others are :-



∴ Three plains for measurement



∴ Horizontal plane

Forensic significance :

- It helps to identify individuals (suspected) if we have the data already.
- It might also help in catching criminals that might have undergone any type of plastic surgery.

- It helps ~~to~~ in forensic examination across diverse population as it will become easy to classify people into groups according to our choice.

14.1

Poirat Parle / Bertillion System :

- Poirat Parle and Bertillion System are the two different methods of measurements but are counted / done as one for better Results.

- Where Poirat Parle focuses on by the facial structure, Bertillion System is followed by the body measurements like ~~Hair~~ Body structure etc.

Comparison with Biometric Methods: [L: Limitation S: Strengths]

Geeta University, Panipat
21 MAY 2025
Asst
Controller of Examination

L These are done on the statement of witness of it may vary (maybe a little or a lot) with actual one, where biometric recognises the exact same.

S As Biometric is not available everywhere, it is best to rely on Portrait Parle / Bertillion system

L It is very time consuming as comparison to how fast biometric methods are.

S Suppose a person tries to trespass the Biometric with fake ~~into~~ fingrprints, then biometric will allow it but in the Bertillion System each and every thing in the body is measured that becomes nearly impossible to copy.

L Facial appearance can be changed by surgery which will make Portrait Parle of no use, but biometric can't be chang like - eyes, fingrpr

Facial Reconstruction :

It is a method / process in which the face of an individual is reconstructed with the help of skull / skull parts for identification.

- It is mainly done in cases like mass disaster or recovering some part of skull from crime scene.

- There are various facial reconstruction techniques like 3-D reconstruction, Image spectroscopy, Video spectroscopy, X-Ray comparison.
- The are mainly categorised in Two parts :
  1) Manual
  2) Digital

∴ Manual :- This include process like image spectroscopy, video spectroscopy, etc.

→ Image : In image spectroscopy the image of the skull is overlapped with the image of the suspected person for identification.

→ Video : It is same as image spectroscopy but the only difference is that we take videos of skull and the person's face for reconstruction.

→ X-Ray : In the the X-Ray of the skull is taken along with the X-Ray of person's head and then matched together

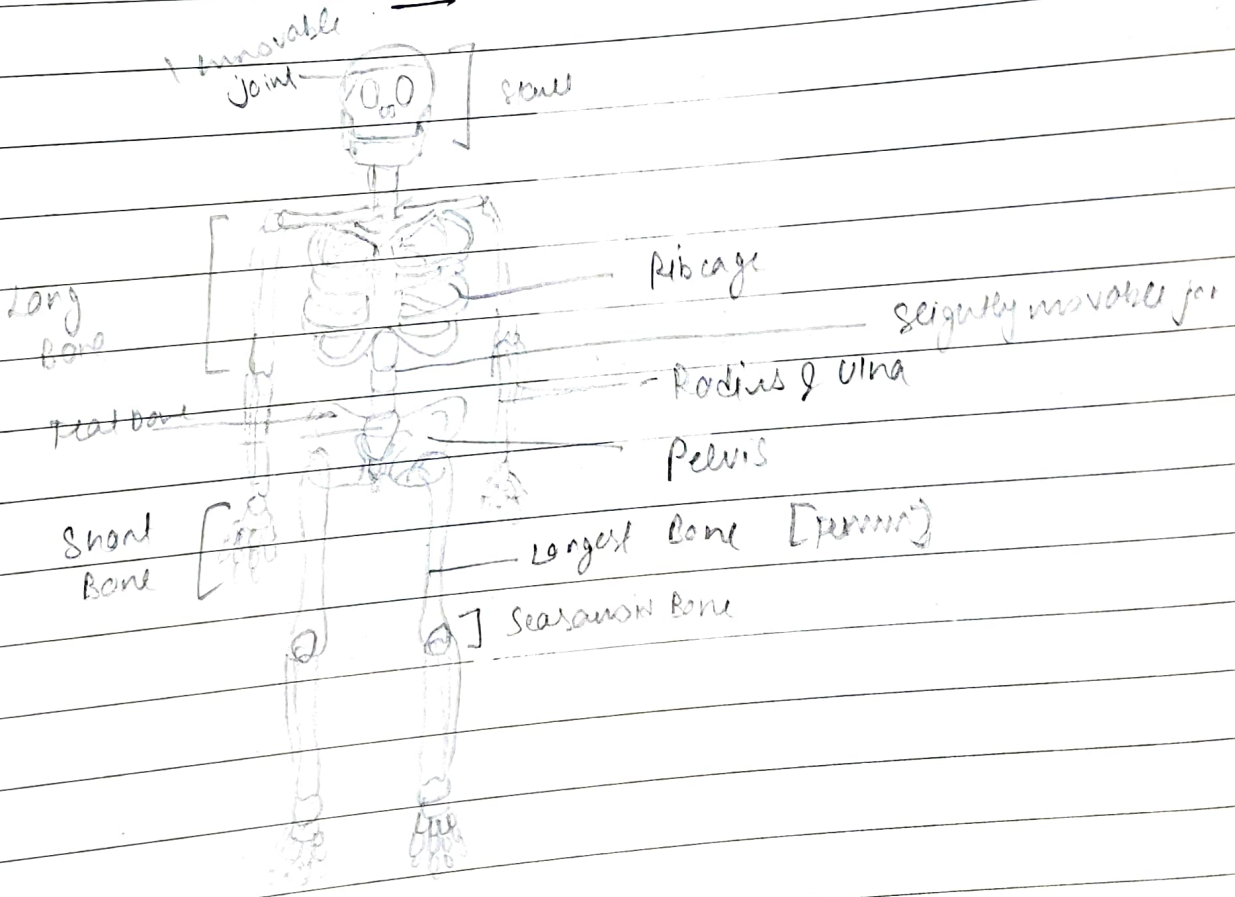
∴ Digital : • Digital methods are more ADVANCED and ACCURATE than manual methods as there might be any miscalculation while overlapping
• It's more reliable.

[Like - 3D reconstruction → where a person's face is created

in our system as a 2D, 3D model

Part-C

16.2



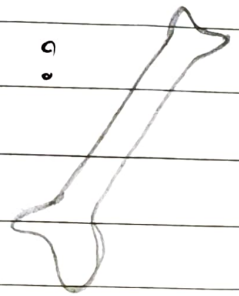
Human Skeleton →

- New born babies usually have 300 - 400 bones
- As a baby grows and becomes older the number of bones get reduced to 206.

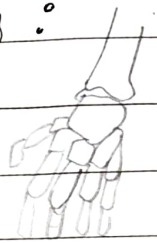
- Main function of human skeleton:
  - Provide structure to the body.
  - Protects vital organs.
  - Supports our body weight.

- There are various types of bones like :-
  - Long Bones
  - Short Bones
  - Flat bones
  - Sesamoid Bones
  - Irregular

(1) Long Bones :



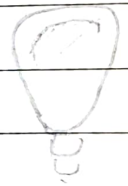
(2) Short Bones :



(3) Irregular bones :



(4) Flat bones :



(5) Sesamoid Bones :

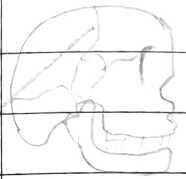


- There are different type of joints in human skelton too.

(1) Immovable (2) Slightly movable (3) Fully movable.

- Immovable

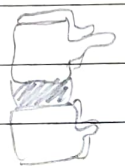
Joints that don't move a bit



(Suture ex)

- Slightly movable

Joints that partially move



- Fully movable

Joints that are freely movable.

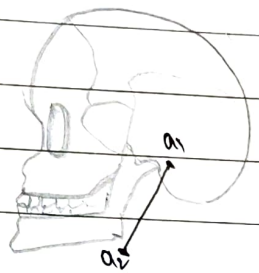


17.2

Sex Determination from Skull :

(8)

- Sex Determination with the help of skull is one of the crucial ways by which we can be helped in identification of an individual. Whether it's a male or female.



∴ Male skull



∴ Female skull

- Larger than female skull
- The muscle attachment is strong
- Mandibular angle is (90°-110°)
- Chin is U-shaped
- Zygomatic bone is more prominent in males

- Small is size than mal skull
- Muscle attachment is weak in comparison to male.
- Mandibular angle is more than 110°
- Chin is V-shaped
- Zygomatic bone is less prominent is females.

• The male skull is more Rugged	• It is less rugger
• Male skull overall appearance is broader than femal skull	• It is comparatively consild than male skull.
• The size of forehead also varies in males and females.	
• A_1 to A_2 is ~~more~~ longer in males	• A_1 to A_2 is shorter in females.
• orbits are Square shaped	• Orbits are rounded

Q.01

(8)

⇒ Forensic odontology: It is the branch of forensic Science that deals with the study of teeth. (mainly of human beings.

Role in Mass Disaster:

→ When there is a mass disaster situation like land slide, Earthquake etc. it becomes very difficult to find clues or evidence for the identification of people.

→ So forensic odontology is opted as one of the ways because as we know Teeth are very strong, so they cannot be easily altered in any way.

- Also with teeth we can also ~~find~~ estimate the age of the individual by means like.

  Total no of Teeth present, the condition of teeth (whether they are of a child, adult or ~~of~~ old age person).

- Dental Records are also used for accurate victim identification when other method fails in a way that:

→ Dental records include whether there is any Anomalii that cannot be fixed completely.

→ ~~Eg:~~ These include – When there is extra teeth
– When there is no teeth
– When ~~the~~ two or more teeth are attached
– When ~~there~~ person had gone any Dental procedure like capping, RCT.

- As these Anamolies are absolute and cannot be completely fixed it becomes easy for the accurate victim identification.

Qx → The victim might not be in any condition for identification in a mass disaster like ~~Land slide~~ burnt forest fire

The face is completely ~~crushed~~ burnt, so we opt for dental records if any as go for the forensic odontology as already mentioned that teeth are too strong to be easily affected.

18.1

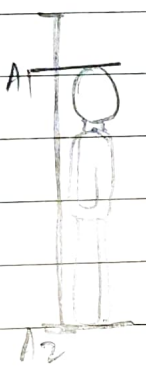
8

- **Somatometry**: It is the method / way of measurement of different body parts like head, face, nose, ear, height, body weight & the purpose of personal identification.

- **Some Somatometric Measurement are**:

(1) **Height**: The measurement should be done when the person is standing straight and still. From head to toe.

eg - From A_1 to A_2

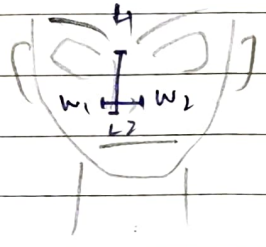


2) Face - For face the measurement is done from one zygomatic bone to the other (called bizygomatic measurement)



eg - From Z_1 to Z_2

3) Nose - As for the measurement of nose two measurement are considered
~~Sen~~ Length and width.



Length L_1 to L_2
width/Breadth W_1 to W_2

Importance in forensic examination:

- The measurement of facial features like nose, ear might help in personal identification

  ex- in portrait parle method and also in facial Reconstruction.

- While there is a diverse population to be examined, it can be easily classified on the basis of

  height and weight as per our own needs.

- Same can with done by collection of data of measurement like bizygomatic measurements etc.

- These measurement can be used for classifying people in different categories making it easy to examine

Reliability and Accuracy:

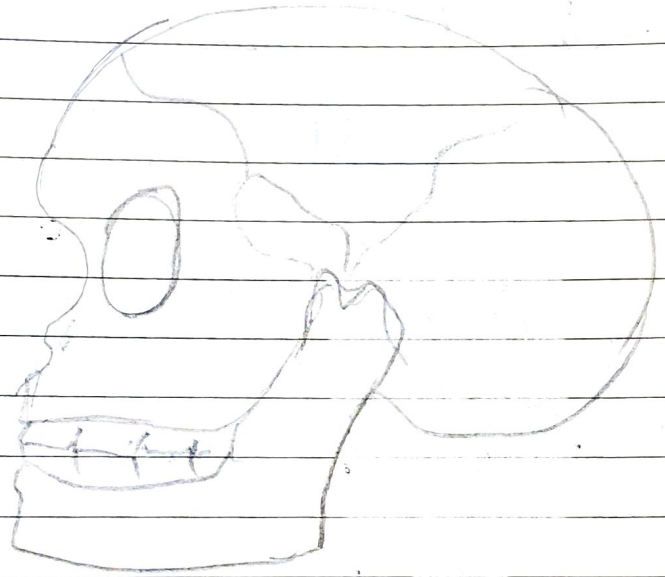
- Some of these might be useful for identification like but some might fail in doing so, like person might have broken a bit or he/she might have change in weight.

- Although it is ~~many impossible but in some case there might be~~ possible that a person might have undergone plastic surgery, changing for measurement of face.

18.1

84

Male Skull:



male skull

- It is more rugged than female.
- The overall size of male skull is bigger than female.
- The muscle attachment to mandible is strong in males.

- The zygomatic bones is more prominent in males.
- Chin shape is "U" in males.
- The mandibular angle is 90°-110° in Males.
- Male skull is more broader than female.
- Mandible is larger than females.
- orbits are square shaped.

19.1

(8)

Facial Superimposition Techniques

There are various Facial superimposition Techniques that include

(1) Image spectroscopy

(2) Video spectroscopy

(3) X-Ray superimposition.

Imag spectroscopy: In this an image of the suspected person is took along with the skull that is recovered and

• Then They both are superimposed on eac

- Video Spectroscopy: In this the video of skull is took along with the video of the suspected person face.
  → And likewise the too are superimposed on each other.
- X-Ray Superimposing: In this they X Ray of Both skull i.e. recovered and the person's X-Ray of face is superimposed on each other.

Superimposition is not only done when recovered full skull it is also done

when we observe a part of skull or mandible.

→ which sometimes may case uncertainity as we are only working on the basis of some part but not on basis of whole skull.

- In confirmed identity case it does not really have any point as we already know the persons identity but in disputed case we need to superimpose a set of images. as the xray, videos as the identity is unknown.

→ They can be used when we have a slight idea of who the person can be.

End of Writing